MILITARY SPECIFICATION

DRIVE, BALLISTICS: M5A2

1. SCOPE

1.1 This specification covers an instrument which is included in a tank's fire control system for the purpose of transmitting superelevation data determined by a ballistics computer (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-F-13926	Fire Control Materiel: General Specification Governing the Manufacture and Inspection of
MIL-P-14232	Parts, Equipment and Tools for Ordnance Materiel, Packaging of
MIL-I-45607	Inspection Equipment, Supply and Maintenance of

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions

DRAWINGS

U.S. Army Munitions Command

F8620200	Drive, Ballistics: M5A2
IEL 8620100	List of Inspection Documents for Drive, Ballistics: M5A2

FSC 1220

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PACKAGING DATA SHEET

MIL-P-14232/P8620100 Packaging of Drives, Ballistics: M5A2, M1O, M1OA1, M10A2, and M1OB1

(Copies of specifications, standards, drawings, and packaging dataa sheets required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Fabrication. - The ballistics drive shall be manufactured in accordance with Drawing F8620200 and all drawings pertaining thereto.

3.1.1 Initial Production sample.- Requirements for the submission of an initial production sample by the contractor shall be as specified in the contract.

3.2 General sPecification,- The contractor shall be responsible for adherence to the following requirements of MIL-F-13926:

(a) Order of precedence
(b) Dimensions and tolerances
(c) Inorganic protective surface finishes
(d) Part identification and marking
(e) Workmanship
(f) Soldering electrical connections

3.3 Environmental service conditions.

3.3.1 Standard ambient temperature.- Unless otherwise specified, the ballistics drive shall meet the requirements of 3.3.3 to 3.4.5 inclusive at a standard ambient temperature between +60° and “+90° Fahrenheit (F).

3.3.2 Extreme operating temPeratures.- The ballistics drive shall meet the requirements of 3.4.4 to 3.4.5 inclusive (except 3.4.4.1, where only the applicable requirements shall apply) while exposed to, and thermally stabilized at, ambient temperatures Of -40° and +150°F9

3.3.3 Vibration.- When mounted and secured in its normal operating position and set at zero gun elevation and zero superelevation, the ballistics drive shall be capable of withstanding a vibration of 30 ~ 5 cycles per second (cps) frequency with an amplitude Of 1/16 inch (118 inch maximum excursion) for a period of 5 minutes ± 15 seconds. Subsequent to vibration the drive shall show no evidence of. damage or physical failure, and it shall meet all the performance requirements of this specification.

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3.4 Performance.

3.4.1 Orientation. - The ballistics drive shall be positioned in accordance with the dimensions shown in Figure 1. (This represents the normal operating position of the drive when mounted in a tank turret.) The index line on the superelevation input shaft shall be aligned to the arrow on the qualification plate for zero superelevation in accordance with the value and direction indicated thereon. The "O" graduation lines on the inner and outer scales of the double eccentric on the 11-inch arm shall be in coincidence with their respective index lines. The stud in Connector Assembly D8201625 shall be secured in such a manner and position simulating normal function with the range finder (see Figure 1); and the "O" graduation lines on the cap and sleeves of the doudle eccentric shall be in coincidence with each other. There shall be at least three fixed positions available at the simulated gun trunnion, representing the following gun elevations:

5 degrees depression (from horizontal)
0 degrees (horizontal)
17 degrees elevation (from horizontal)

3.4.2 SuPerelevation.

3.4.2.1 Input travel.- The input shaft shall be rotatable clockwise and counterclockwise. The limit of travel shall be effected wholly by the stops on the split nut and extremes of the lead screw. There shall be no interference, jamming, or torque build-up as a result of approaching or reaching the stops. A minimum of 30 full turns of the input shaft shall be available over the entire range of rotation.

3.4.2.2 Accuracy.- With the n-inch arm in a fixed position and the ballistics drive oriented in accordance with 3.4.1 and Figure 1, backlash excluded, and with the 200 inch pound load aPPlied to the assembly the ballistics drive shall meet the superelevation accuracy specified below:

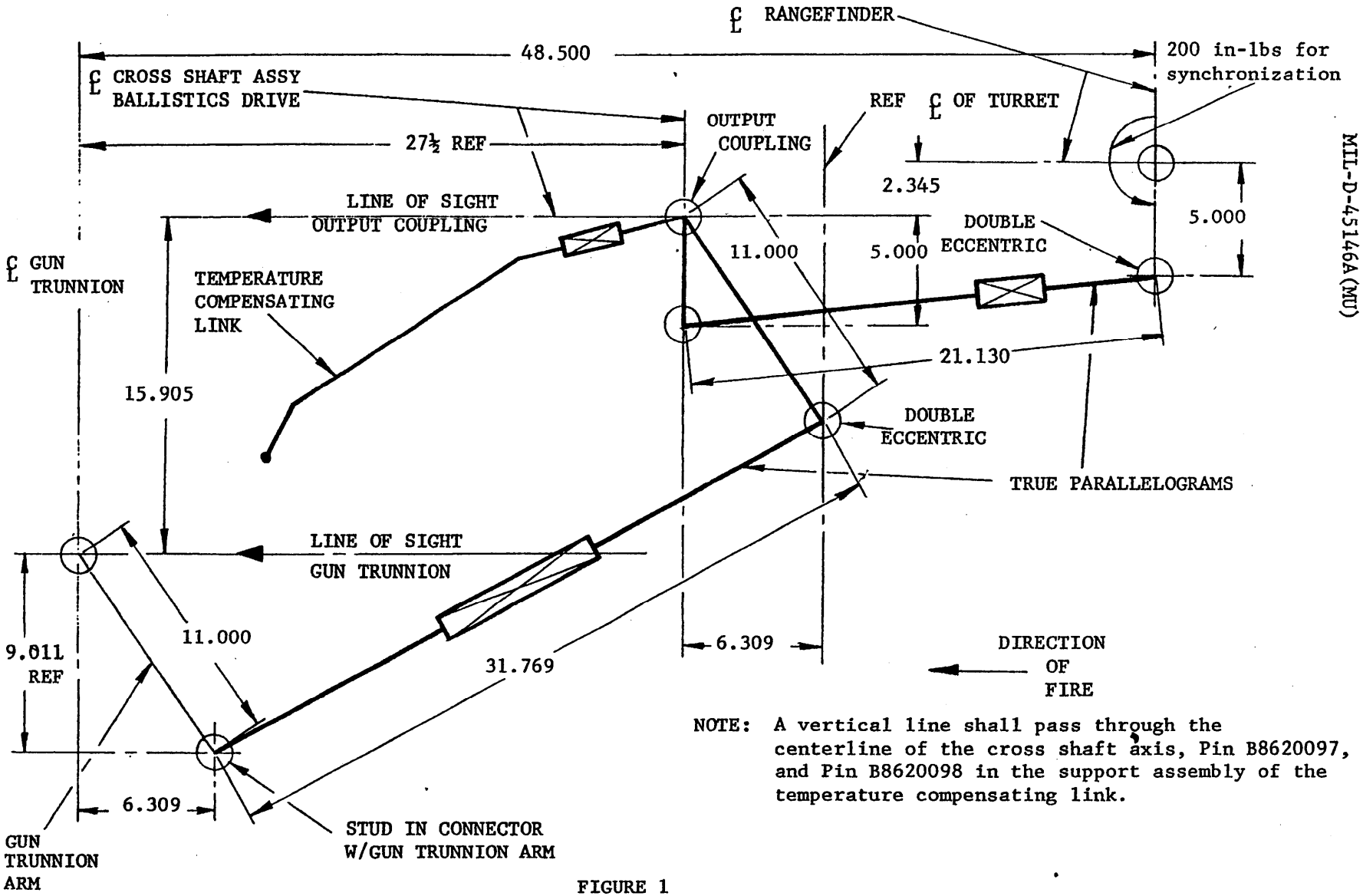
Input Shaft Rotation* (Complete revolutions)	Output Coupling Rotation	Tolerance
5 - Clockwise	24.95 mils	+0.10 mil
10 - Clockwise -	49.94 mils	± 0,10 mil

*origin of input. shaft rotation at Box Assembly F8201640.

3.4.2.3 Backlash.- With no load applied to the assembly, backlash in the superelevation mechanism shall not exceed 0.1 mil as a result of change in. rotation direction of the superelevation input shaft. Backlash shall be checked at the following positions of the superelevation input shaft:

(a) Zero superelevation.
(b) Five (5) complete clockwise revolutions of - the input shaft from the zero superelevation setting.

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NOTE: A vertical line shall pass through the centerline of the cross shaft axis, Pin B8620097, and Pin B8620098 in the support assembly of the temperature compensating link.

FIGURE 1

NOMINAL POSITIONING OF BALLISTICS DRIVE M5A2 ASSEMBLED IN TURRET

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(c) Ten (10) complete clockwise revolutions of the input shaft from the zero superelevation setting.

3.4.3 Synchronization.

3.4.3.1 Conditions for synchronization.

(a) A constant torque of 200 pound inches applied as indicated in Figure 1.
(b) The line of sight at the output coupling parallel to the simulated gun trunnion line of sight, with the ballistics drive positioned at zero gun elevation in accordance with 3.4.1 and Figure 1.
(c) All planetary error due to superelevation input shaft rotation deleted.

3.4.3.2 Accuracy.- The line of sight established at the output coupling (simulating the periscope line of sight) shall not deviate from the line of sight established at the simulated gun trunnion by more than 0.2 roil. This requirement shall be met by. approaching each of the Simulated gun positions specified in 3.4.1 from low to high, and shall again meet the requirement at each position going from high to low.

3.4.3.3 Backlash.- The difference between the values recorded as a result of elevation and depression at each of the above simulated gun positions shall not indicate backlash in excess of 0.1 mil.

3.4.4 operability.

3.4.4.1 Rotating studs.- The torque required to rotate the studs located at the free end of the linkages which connect to the simulated gun trunnion and rangefinder shall be 3.0 ± 0.5 pound inches at the standard ambient temperature of 3.3.1. The torque required to rotate the studs at the extreme operating temperatures of 3,3.2 shall not exceed the value obtained at standard ambient temperature by more than one pound inch.

3.4.4.2 Superelevation input shaft.- The torque required to rotate the superelevation input shaft (at its point of origin) throughout the entire range of travel clockwise and counterclockwise shall not exceed 4 pound inches at the standard ambient temperature of 3.3.1 and the extreme operating temperatures of 3.3.2.

3.4.4.3 Level vial cover.- The level vial cover shall .rotate freely and be held firmly in both the open and closed positions when the detent is engaged. This requirement shall be met at the standard ambient temperature of 3.3.1 and the extreme operating temperatures of 3.3.2.

3.4.5 Light assembly.- The lamp shall light when it is energized by b 3-volt connected to the receptacle. This requirement shall be met at the standard ambient temperature of 3.3.1 and the extreme operating temperatures of 3.3.2.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for insPection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. . Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General Provisions.- The component and subassembly inspection requirements of MIL-F-13926 form a part of the Quality Assurance Provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

4.2 Initial Production test.- Initial production test samples shall be submitted in accordance with contract requirements, Unless otherwise specified in the contract, samples shall be the first two production units and shall have been manufactured using the same materials, processes, and procedures as will be used in regular production. Initial production testing shall be in accordance with 4.3, 4.4, and 4.6 and shall be witnessed by the Government representative for all examinations and tests. The Government reserves the right to conduct any or all examinations or tests as may be deemed necessary.

4s2.1 Initial Production sample failure.- Should one unit fail to meet the initial production testing specified in 4.2, the contractor shall cease those operations which are causing the rejection and make the necessary corrections thereto. When corrections in the process or controls have been made, two additional (new) production models shall be submitted in accordance with 4.2,

4.3 Inspection provisions.

4.3.1 Submission of Product.- Unless otherwise specified by the Contracting Officer, inspection lot size, lot formation, and presentation of lots shall be in accordance with “Submission of Product” provisions of MIL-STD-105.

4.3.2 Examination and tests.- Components and subassemblies shall be inspected in accordance with the inspection provisions contained in MIL-F-13926, and the Supplementary Quality Assurance Provisions (SQAP) listed in the List of Inspection Documents, IEL 8620100. Examinations and tests related to Section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with MIL-STD-105, and the sampling plans specified in Tables I, II, and III herein. Examinations and tests for packaging, packing, and marking shall be in accordance with MIL-P-14232 and Section 5 herein. The tabulated classification of defects shall constitute the minimum inspection to be performed by the supplier prior to Government acceptance or rejection by lot.

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TABLE I - CLASSIF1CATION OF DEFECTS

CLASS	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
CRITICAL: NONE			
MAJOR : AQL 1.5% Defective			
101.	Vibration	3.3.3	4.6:2
102.	Superelevation, input travel	3.4.2.1	4.6.3.1
103.	Superelevation, accuracy	3.4.2.2	4.6.3.2
104.	Superelevation, backlash	3.4.2.3	4.6.3.3
105.	Synchronization, accuracy	3.4.3.2	4.6.3.4
106.	Synchronization, backlash	3.4.3.3	4.6.3,5
107.	Operability, rotating studs (60° to 90°F) ‘	3.4.4.1	4.6.4.1
108.	Operability, superelevation input shaft (60° to 90°F)	3.4.4.2	4.6.4.2
109.	Operability, level vial cover	3.4.4.3	4.6.4.3
110.	Light assembly (continuity)	3.4.5	4.6.4.4
MINOR: NONE			

4.3.3 Acceptance and rejection.- Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with “Acceptance and Rejection” as specified in MIL-STD-105.

4.4 Special sampling.

4.4.1 General.- One ballistics drive shall be selected at random as a special sampling from each 100 produced. The sample shall meet the requirements and tests in Table 11.

TABLE II

	CHAWCTERISTIC	REQUIREMENT	TEST. PROCEDURE
301.	Fabrication	3.1	Applicable drawings - visual
302.	General specifications	3.2	MIL-F-13926 - visual

4.4.2 Environmental.- Three units shall be selected at randomby a Government representative as special samples from each 50 produced or from each month’s production, whichever occurs first. The samples shall have met the requirements and tests in Table I and shall then meet the requirements and tests in “Table III.

TABLE III

	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
303.	Extreme operating temp. (-40° and +150°F)	3.3.2	4.6.1
304.	Operability rotating studs (-40° and +-150°F)	3,4.4.1	4.6.4.1
305.	Operability superelevation input shaft (-40° and +150°F)	3.4.4.2	4.6.4.2

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4.4.3 Failure of sample.- Should any one item of a special sampling fail to meet the specified test requirements, acceptance of the product shall be suspended by the Government until necessary corrections have been made by the contractor,and the resubmitted samples have been accepted (see 4.3.3).

4.5 Inspection equipment.- Except as otherwise provided for by the contract, the contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-I-45607.

4.5.1 Government furnished inspection equipment.- Where the contract provides for Government furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirerments of MIL-I-45607.

4.5.2 Contractor furnished inspection equipment.

4.5.2.1 Government design.- Unless otherwise specified in the contract, all inspection equipment specified by drawing number in specifications or SQAP forming a part of the contract, shall be supplied by the contractor in accordance with technical data listed in IEL 8620100.

4.5.2.2 Contractor design.- The contractor shall design and supply inspection equipment compatible with “Test Methods and procedures" specified in 4.6 and with the component inspection procedures specified in “Examination” and “Test facilities" requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within ten percent of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions, and tolerances used in the design of test equipment shall be so selected and controlled as to insure that the test equipment will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limtt and permit positive rejection when non-conforming. Construction shall be such as to facilitate routine calibration of the test equipment,

4.5.2.3 Inspection equipment design requirements.

4.5.2.3.1 Special testing equipment.- Special testing equipment utilized for tests herein shall conform to Drawing F8227525, Fixture, Final Inspection for Drive, Ballistics: M5A2

4.6 Test Methods & Procedures.

4.6.1 Extreme operating temperatures- Testing equipment utilized in this test shall be in accordance with the “Test Facilities” requirements of MIL-F-13926 and the conditions of 3.3.2. The ballistics drive shall be exposed to each temperature specified in 3.3.2 for a period of five-hours. At the end of each five hour period, and while still at the temperature extreme, the ballistics drive shall be subjected to the tests specified in 4.6.4.1 and 4.6.4.2.

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4.6.2 Vibration.- Standard vibrating equipment capable of providing the amplitude and frequency specified in 3.3.3 shall be utilized for the vibration test. Unless otherwise specified, the test equipment shall be in accordance with the "Test Facilities requirements of MIL"-F-13926. The vibrating equipment shall be equipped with a suitable adapter to permit mounting and securing the ballistics drive in the normal operatin position. The input and output shafts shall be immobilized. The ballistics drive shall be mounted on the standard vibrating equipment specified. The motion shall be in a vertical plane for a period of five minutes plus or minus fifteen seconds. Subsequent to vibration, the ballistics drive shall be thoroughly examined for physical defects resulting from inferior materials or poor workmanship, to determine compliance with 3.3.3, prior to being subjected to the remaining tests specified herein. A visual and tactile examination shall be conducted to check functioning of moving parts for undue irregularities of movement, friction, or looseness. Ballistics drives which have passed the above examination shall be subjected to the tests of 4.6.3 to 4.6.3.5 inclusive to determine overall compliance with 3.3.3.

4,6.3 Performance.- The following tests shall be performed with the ballistics drive positioned on the special testing equipment specified in 4.5.2.3.l. Setup of the test equipment and installation of the ballistics drive under test shall be performed in accordance with the setup procedures which are included on the test equipment drawing. Subsequent to installation, the ballistics drive shall be positioned and oriented in the following manner to conform to the requirements of 3.4.

(a) Sight through auto collimator "A" to mirror block and autocollimate to the zero degree mirror by adjusting the trunnion arm of the test equipment. This is zero degree gun elevation. Note the position of the level bubble in the level vial and the alignment of the zero graduation lines on the inner and outer scales of the double eccentrics. The bubble in the level vial shall center within the graduation lines, and the zero graduation lines on the inner and outer scales of the double eccentric shall be in coincidence with their respective index lines as required by 3.4.1.

(b) Apply a 200 inch pound load on the cross shaft by mounting a 40 pound weight to the torque wheel assembly of the test equipment. With the stop pins retracted, attached the appropriate clamp block screw, or adapter assembly, to the junction box input coupling of the ballistics drive under test.

(c) Rotate the input shaft in a counterclockwise direction by means of the clamp block screw or ths adapter assembly, until the" travel stop in the superelevation box is engaged. From this position rotate the input shaft in a clockwise direction the required number of turns indicated on the qualifying plate, minus one turn. Sight through

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locating pin of the clamp block screw, or the adapter assembly, between the appropriate stop pins. This is 0,0 degree superelevation. Note the alignment of the index line on the input shaft with the arrow on the qualifying plate and record the complete number of clockwise turns the input shaft was rotated, Failure of the index line to be aligned with the arrow in accordance with the value and direction indicated on the qualifying plate shall be cause for rejection of the ballistics drive under test.

4.6.3.1 SuPerelevation, input travel.- Retract the stop pins of the clamp block screw, or the adapter assembly, to allow the input shaft to rotate freely during this test. Rotate the input shaft from the counterclockwise travel stop to the clockwise travel stop, then back to the counterclockwise travel stop. During this procedure the number of turns of the output shaft shall be counted in one direction. Travel of the input shaft shall conform to the requirements of 3.4.2.1, Subsequent to this test return the input shaft to the zero degree superelevation position, in accordance with the procedure specified in 4.6.3(c).

NOTE : The gun trunnion arm shall be immobilized at zero degree gun elevation during this test (4.6.3.1) and for the following tests specified in 4.6.3.2 and 4.6.3.3.

4.6.3.2 SuPerelevation accuracy.- Retract the stop pins of the clamp block screw, or the adapter assembly. With backlash removed for clockwise travel, rotate the input shaft 5 complete revolutions clockwise from zero degree superelevation position, stopping on the fifth revolution at the appropriate, extended stop pin. Sight through autocollimator "B" to the correct mirror of mirror block "C". Record the position of the boresight point of the reflected image, as measured on the autocollimator reticle. Retract the stop pin, Rotate the input shaft 5 additional revolutions in the same direction (clockwise a total " of 10 revolutions from the zero degree superelevation starting point) stopping on the tenth turn by extending the correct stop pin. Sight through autocollimator "B" to the correct mirror of mirror block "C". Record the position of the boresight point of the reflected image, as measured on the autocollimator reticle. The rotational position of the output coupling, as determined by the recorded values, shall be within the tolerance specified in 3.4.2.2.

4.6.3.3 Superelevation, backlash.- The test equipment specified in 4,6.3 shall be utilized for this test, except the 40 pound weight on the torque wheel shall be removed. Perform this test in conjunction with 4.6.3.2.with backlash removed"for.clockwise. travel. " Position the input shaft at zero degrees superelevation as specified in 4.6.3(c), and record this reading. Observe the index line on the input shaft and the arrow on the qualifying plate. The index line and the arrow on the qualifying plate will not be aligned, due to the removal of the 40 pound weight from the torque wheel assembly. Failure of the index line to be aligned with the arrow shall not be cause for rejection when this test is performed. Retract the stop pins of the clamp block screw, or the adapter assembly. With the backlash excluded, rotate the input shaft one

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additional turn clockwise, then reverse direction (with backlash included) and stop at the zero position designated by the correct stop pin. Sight through the autocollimator and record the position of the boresight point (backlash) of the reflected image, as measured on the autocollimator reticle. Retract the stop pin. With backlash excluded, rotate the input shaft five complete revolutions clockwise from the zero degree superelevation position, stopping on the fifth revolution by extending the appropriate stop pin. Sight through the autocollimator and record the position of the boresight point of the reflected image, as measured on the autocollimator reticle. Retract the stop pin. With backlash excluded, rotate the input shaft one additional turn clockwise, then reverse rotation of the input shaft (with backlash included) and stop on the completion of the fifth revolution by extending the appropriate stop pin. Sight through the autocollimator and record the position of the boresight point of the reflected image, as measured on the autocollimator reticle. Again retract the stop pin. With backlash excluded, rotate the input shaft five additional complete revolutions clockwise (ten total from zero degree superelevation position), stopping on the completion of the tenth turn by extending the appropriate stop pin. Sight through the autocollimator and record the position of the boresight point of the reflected image, as measured on the autocollimator reticle. Retract the stop pin. With backlash excluded, rotate the input 'shaft one additional turn clockwise, then reverse the direction of the shaft (backlash included) and stop at the tenth turn position by extending the stop pin. Sight through the autocollimator and record the position of the boresight point of the reflected image, as measured on the autocollimator reticle. Any difference between the recorded readings taken at zero, fivej and ten degree superelevation clockwise rotation of the input shaft is the amount of backlash present. Backlash resulting from rotation at any one of the three positions shall not exceed the tolerance specified in 3.4.2.3.

4.6.3.4 Synchronization accuracy. This test shall be performed with the Shaft 8200797 or 8201837 disassembled from the Universal Assembly 8200862. This eliminates any planetary error due to superelevation input shaft rotation. The testing equipment specified in 4.6.3 shall be utilized for this test, except the mirror block “C” is not required. Mirror block “B[t]’ shall be mounted in position and used in place of mirror block “C”. The gun trunnion arm shall be positioned at 0.0 degree gun elevation and the 40 pound weight shall be mounted to the torque wheel assembly of the test equipment. Sight through autocollimator "B" to the appropriate mirror of mirror block “B” and establish 0.0 degree superelevation, utilizing the procedure specified in 4.6.3(c). Without disturbing the 0.0 degree superelevation position, disassemble the shaft coupling from the coupling of the Universal Assembly by removing 4 screws, Prior to disassembly, the exact position of the Universal in relation to the assembly shall be established by appropriate reference marks, as it is necessary to return it to its original position when the specified tests have been completed. Subsequent to disassembly, sight through the autocollimator "B" and reestablish the 0.0 degree superelevation, providing autocollimation to the appropriate mirror of mirror block “B[1]’ has not been maintained. Do not disturb, nor make any further adjustments to the

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Universal assembly for the remainder of the test. The following movements in elevation and depression shall be induced to the gun trunnion arm and autocollimator "A" by actuating the trunnion arm of the test equipment. Position the gun trunnion arm and autocollimator "A" at an angle greater than minus five degrees in depression. With backlash excluded, elevate the gun trunnion arm and autocollimator "A" until" autocollimation is established to the minus five degree mirror. Sight through autocollimator "B" and record the position of the boresight point of the reflected image, as measured on the autocollimator. Elevate the gun trunnion arm and autocollimator "A" (without backlash) until autocollimation is established to the zero degree mirror. Sight through autocollimator "B'[f] and record the position of the boresight point of the reflected image, as measured on the autocollimator. Elevate the gun trunnion arm and autocollimator "A" (without backlash) until autocollimation is established to the plus seventeen degree mirror. Sight through autocollimator "B" and record the position of the boresight point of the reflected image, as measured on the autocollimator. Elevate beyond the 17 degree mirror and eliminate backlash. Depress the gun trunnion arm and autocollimator "A" until autocollimation is established to the plus 17 degree mirror. Record this setting. Depress the gun trunnion arm and autocollimator (without backlash) until autocollimation is established to the zero degree mirror. Sight through autocollimator "B" and record the position of the boresight point of the reflected image, as measured on the autocollimator. Depress the gun trunnion arm and autocollimator "A" (without backlash) until autocollimation is established to the minus five degree mirror. Sight through autocollimator "B" and record the position of the boresight point of the reflected image as measured on the autocollimator. At each of the recorded positions, the ballistics drive shall conform to the accuracy requirements of 3.4.3.2.

4.6.3.5 Synchronization, backlash.- Perform this test in conjunction with the test in 4.6.3.4, The difference in the two recorded readings (elevation and depression) at each position of zero, -5, and +17 degrees in 4.6.3,4 shall represent the amount of backlash resulting from change in direction during rotation of the input shaft. This difference shall not exceed the limit specified for backlash in 3.4.3.3.

4.6.4 Operability. - Prior to conducting the following operability tests, adjust the gun trunnion arm and autocollimator "A" until autocollimation is established to the zero degree mirror. While utilizing" the reference marks made prior to disassembly, secure the coupling of the universal assembly and the shaft coupling in their original position.

4.6.4.1 Rotating studs,- Standard torque measuring equipment capable of measuring torque values of at least five inch pounds shall be utilized for this test. An appropriate, shown on Drawings B8636458, B8636459, and B8636460, to accommodate the rotating studs, shall be used with the torque measuring equipment. Torque Shall be measured while rotating the studs through the full range of travel. The running torque required to rotate the studs shall not exceed the values specified in 3.4.4.1 for the temperature specified in 3.3.1 and 3.3.2.

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4.6.4.2 Superelevation input shaft*- The torque measuring equipment used in 4.6.4.1 shall be utilized for this test except that an appropriate adapter to accommodate the superelevation input shaft (at its point of origin) shall be used. Torque shall be measured while rotating the superelevation input shaft throughout the entire range of travel in the clockwise and counterclockwise directions. The running torque required to rotate the superelevation input shaft shall not exceed the tolerance specified in 3.4.4.2 for the temperatures specified in 3.3.1 and 3.3.2.

4.6.4.3 Level vial cover.- The level vial cover shall be manually operable under the conditions specified in 3.4.4.3. The cover shall rotate freely, with the inspector's hands protected by heavy winter gear or gloves during the extreme cold (minus) temperature test (3.3.2) and it shall not bind during the extreme hot (plus) temperature test (3.3.2) when the detent is engaged at the open and closed positions.

4.6.4.4 Light assembly.- The elevation quadrant light shall light when energized by a three volt source when attached to the receptacle as required by 3.4.5 and shall be operable at the extreme temperatures required by 3.3.2.

5. PREPARATION FOR DELIVERY

5.1 Packing. Packing. and marking.- Packaging, packing, and marking shall be in accordance with MIL-P-14232 and Packaging Data Sheet MIL-P-14232/P8620100. The level of protection shall be as specified in the procurement document.

6. NOTES

6.1 Intended use.- Ballistics Drive: M5A2 is a differential drive which transmits superelevation data received from a ballistics computer and causes an angular displacement of the direct linkage to the rangefinder and the coupling to the gunnerls periscope. This depresses the lines of sight of the rangefinder and the periscope the correct number of roils of superelevation. Without change in superelevation, the ballistics drive acts as a solid linkage providing equal angular movement of the lines of sight of the rangefinder and gunner's periscope corresponding to the movement of the gun in elevation or depression.

6.1.1 The ballistics drive is intended for use in the fire control system of 90-mm gun tanks M48A2 and M48A2C.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Selection of applicable levels of preservation, packaging, and packing.
(c) Applicable stock number.

Custodian:
Amy - MU

Preparing activity:
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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

DD FORM 1 OCT 64 1426

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DEPARTMENT OF THE ARMY

Frankford Arsenal
Philadelphia, Pa. 19137

OFFICIAL BUSINESS

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DEPARTMENT OF THE ARMY

Commanding Officer
Frankford Arsenal
ATTN: SMUFA- N1100
Philadelphia, Pa. 19137

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